

Homework 6: R Functions

Daniel Kim

```
library(bio3d)
```

```
s1 <- read.pdb("4AKE") # kinase with drug
```

Note: Accessing on-line PDB file

```
s2 <- read.pdb("1AKE") # kinase no drug
```

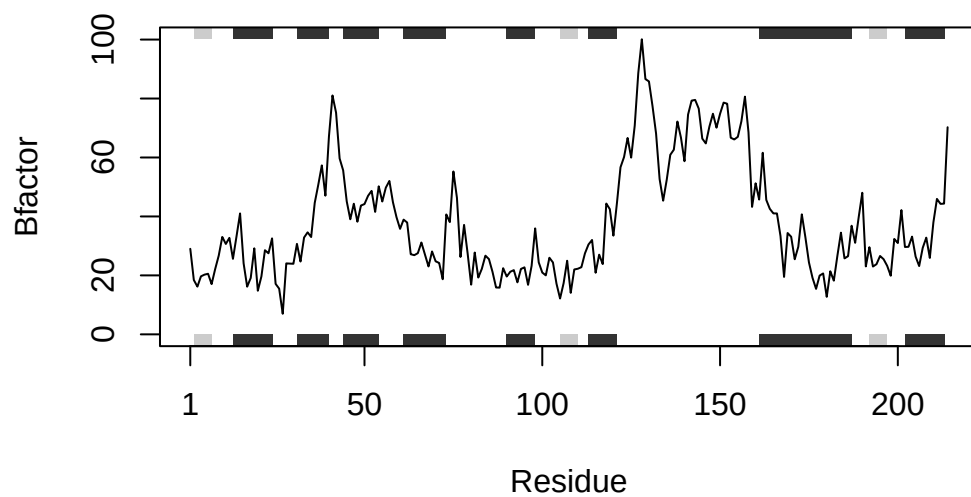
Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

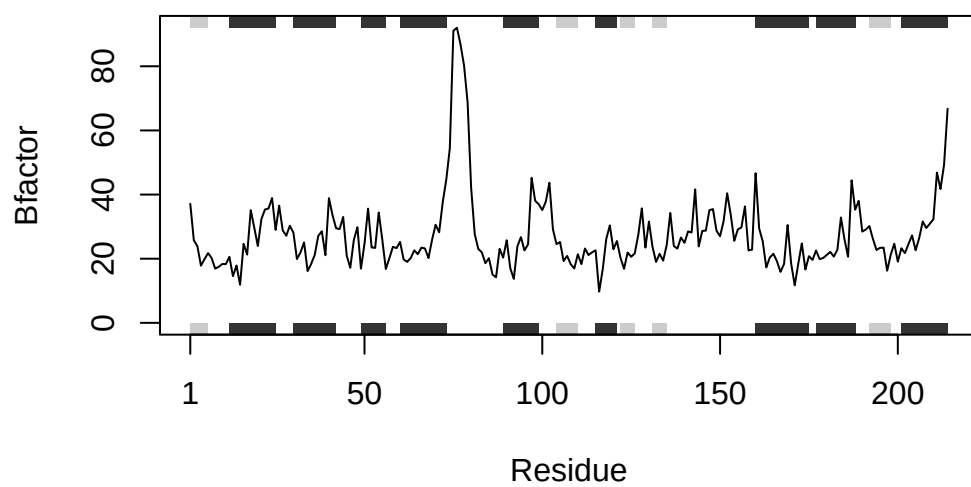
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

Note: Accessing on-line PDB file

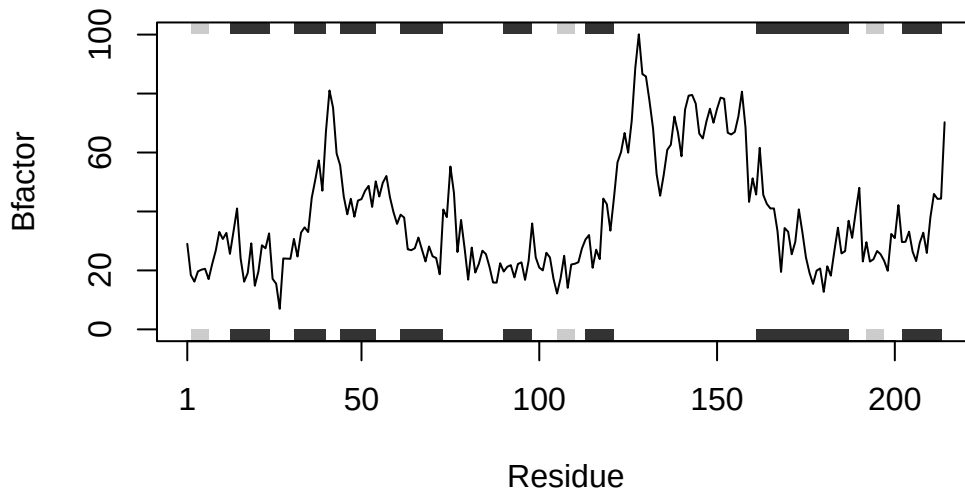
```
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")
s3.chainA <- trim.pdb(s1, chain="A", elety="CA")
s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



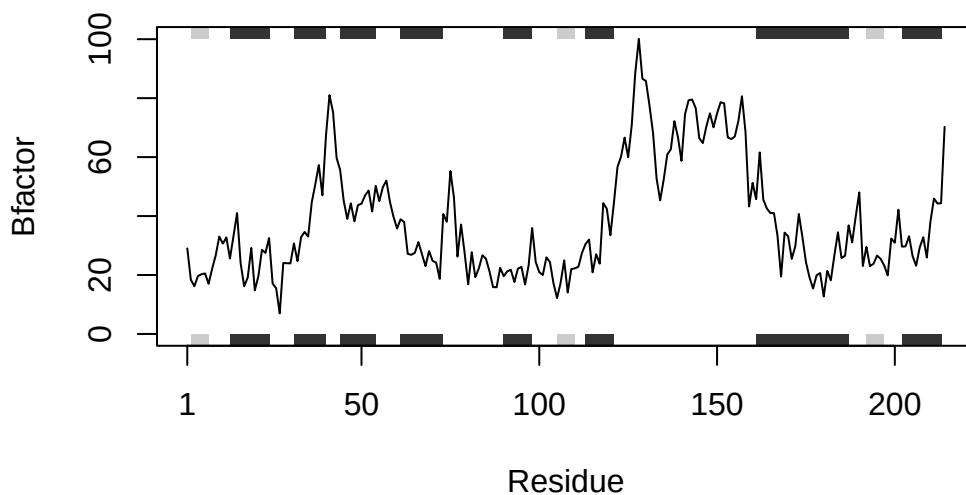
```
#the function takes in the pdb_id as well as a default chain = A arguments
#the function takes in the arguments and plots the bfactors against the trimmed in a line graph
analyze_pdb <- function(pdb_id, chain="A") {
  #reads pdb file from the id
  pdb <- read.pdb(pdb_id)
  #performs the trim operation with the chain as A and elety as CA
  trimmed <- trim.pdb(pdb, chain=chain, elety="CA")
  #extracts bfactors
  bfactors <- trimmed$atom$b
  #plots bfactors against trimmed, using line graph
  plotb3(bfactors, sse=trimmed, typ="l", ylab="Bfactor")
  #the function returns a numeric vector bfactors
  return (bfactors)
}

analyze_pdb("4AKE")
```

Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):

/var/folders/d4/59bk_tmx567frp1cqwr5gwgw0000gn/T//RtmpWCv7Tg/4AKE.pdb exists.
 Skipping download



[1]	29.02	18.44	16.20	19.67	20.26	20.55	17.05	22.13	26.71	33.05
[11]	30.66	32.73	25.61	33.19	41.03	24.09	16.18	19.14	29.19	14.79
[21]	19.63	28.54	27.49	32.56	17.13	15.50	6.98	24.07	24.00	23.94
[31]	30.70	24.70	32.84	34.60	33.01	44.60	50.74	57.32	47.04	67.13
[41]	81.04	75.20	59.68	55.63	45.12	39.04	44.31	38.21	43.70	44.19
[51]	47.00	48.67	41.54	50.22	45.07	49.77	52.04	44.82	39.75	35.79
[61]	38.92	37.93	27.18	26.86	27.53	31.16	27.08	23.03	28.12	24.78
[71]	24.22	18.69	40.67	38.08	55.26	46.29	26.25	37.14	27.50	16.86
[81]	27.76	19.27	22.22	26.70	25.52	21.22	15.90	15.84	22.44	19.61
[91]	21.23	21.79	17.64	22.19	22.73	16.80	23.25	35.95	24.42	20.96
[101]	20.00	25.99	24.39	17.19	12.16	17.35	24.97	14.08	22.01	22.26
[111]	22.78	27.47	30.49	32.02	20.90	27.03	23.84	44.37	42.47	33.48
[121]	44.56	56.67	60.18	66.62	59.95	70.81	88.63	100.11	86.60	85.80
[131]	77.48	68.13	52.66	45.34	52.43	60.90	62.64	72.19	66.75	58.73
[141]	74.57	79.29	79.53	76.58	66.40	64.76	70.48	74.84	70.11	74.82
[151]	78.61	78.24	66.70	66.10	67.01	72.28	80.64	68.54	43.23	51.24
[161]	45.72	61.60	45.61	42.57	41.03	41.02	33.34	19.48	34.38	33.11
[171]	25.48	29.68	40.71	32.91	24.41	19.20	15.43	19.93	20.66	12.72
[181]	21.40	18.21	26.68	34.50	25.77	26.52	36.85	31.05	39.84	48.03

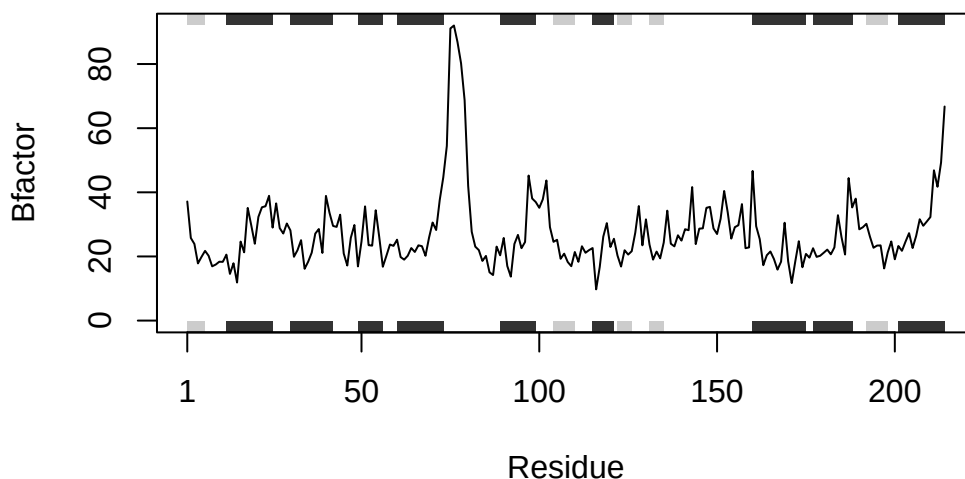
[191]	23.04	29.57	23.00	23.80	26.59	25.49	23.25	19.89	32.37	30.97
[201]	42.16	29.64	29.69	33.15	26.38	23.17	29.35	32.80	25.92	38.01
[211]	45.95	44.26	44.35	70.26						

```
analyze_pdb("1AKE")
```

Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
 /var/folders/d4/59bk_tmx567frp1cqwr5gwgw0000gn/T//RtmpWCv7Tg/1AKE.pdb exists.
 Skipping download

PDB has ALT records, taking A only, rm.alt=TRUE



[1]	37.14	25.76	23.90	17.83	19.86	21.75	20.21	16.92	17.47	18.35	18.31	20.57
[13]	14.56	17.87	11.87	24.63	21.29	35.13	29.68	23.96	32.34	35.34	35.64	38.91
[25]	29.00	36.55	28.83	27.15	30.28	28.13	19.90	21.95	25.07	16.15	18.35	21.19
[37]	27.13	28.55	21.10	38.88	33.63	29.51	29.21	33.01	20.92	17.17	25.84	29.80
[49]	16.89	24.66	35.62	23.52	23.37	34.41	25.96	16.79	20.20	23.72	23.29	25.23
[61]	19.81	19.00	20.21	22.62	21.40	23.47	23.20	20.21	25.90	30.58	28.25	37.60
[73]	44.66	54.46	91.10	92.02	86.85	80.21	68.72	42.01	27.69	23.06	21.98	18.60

```

[85] 20.17 15.06 14.20 23.07 20.36 25.76 17.02 13.71 23.88 26.72 22.58 24.51
[97] 45.23 38.07 36.97 35.17 37.83 43.69 29.14 24.56 25.20 19.27 20.88 18.27
[109] 16.96 21.38 18.33 23.18 21.15 21.97 22.63 9.74 16.71 26.18 30.39 22.95
[121] 25.51 20.28 16.86 21.94 20.59 21.64 27.42 35.72 23.47 31.57 23.71 19.01
[133] 21.52 19.40 24.32 34.28 23.96 23.14 26.60 24.94 28.49 28.18 41.64 23.85
[145] 28.67 28.76 35.16 35.46 28.74 26.99 31.74 40.41 33.73 25.57 29.13 29.74
[157] 36.32 22.58 22.82 46.67 29.44 25.40 17.27 20.38 21.55 19.19 15.89 18.37
[169] 30.51 18.47 11.70 18.45 24.75 16.63 20.80 19.62 22.56 19.87 20.22 21.16
[181] 22.13 20.66 22.82 32.86 26.04 20.60 44.44 35.28 38.03 28.46 29.10 30.19
[193] 26.17 22.71 23.39 23.44 16.27 21.26 24.67 19.12 23.26 21.75 24.59 27.26
[205] 22.63 26.40 31.60 29.57 30.90 32.29 46.86 41.73 49.31 66.76

```

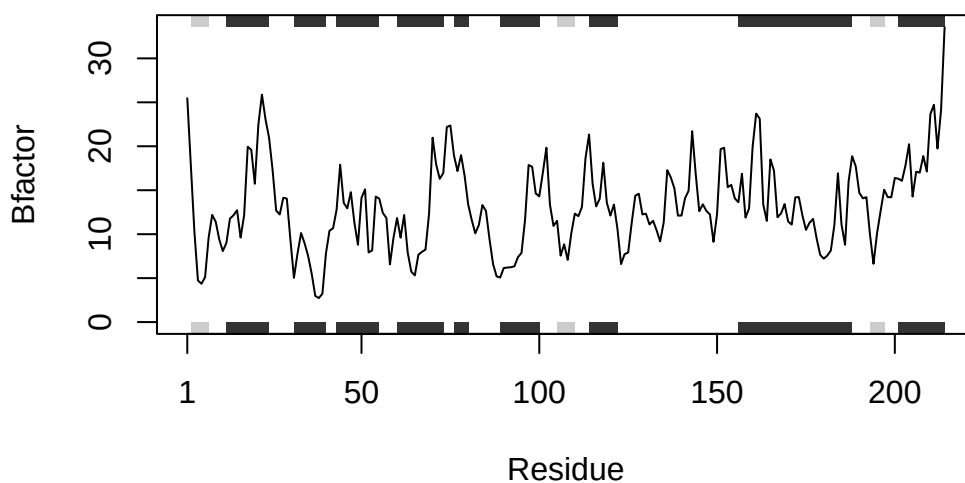
```
analyze_pdb("1E4Y")
```

Note: Accessing on-line PDB file

```

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/d4/59bk_tmx567frp1cqwr5gwgw0000gn/T//RtmpWCv7Tg/1E4Y.pdb exists.
Skipping download

```



```
[1] 25.46 17.86 10.28 4.73 4.36 5.10 9.59 12.19 11.41 9.39 8.08 9.01
```

[13]	11.77	12.15	12.72	9.62	12.18	19.95	19.59	15.73	22.51	25.87	23.08	20.97
[25]	17.28	12.69	12.24	14.14	14.05	9.38	5.03	7.78	10.13	8.96	7.50	5.48
[37]	2.97	2.73	3.23	7.81	10.40	10.67	12.79	17.90	13.56	12.94	14.78	11.31
[49]	8.79	14.13	15.10	7.92	8.15	14.28	14.04	12.42	11.84	6.57	9.59	11.84
[61]	9.61	12.18	7.89	5.74	5.31	7.67	7.99	8.24	12.34	20.98	17.93	16.30
[73]	16.94	22.19	22.36	18.96	17.18	18.99	16.65	13.39	11.61	10.10	11.03	13.31
[85]	12.66	9.44	6.60	5.20	5.06	6.16	6.20	6.24	6.34	7.39	7.86	11.66
[97]	17.87	17.67	14.63	14.30	16.98	19.84	13.36	10.93	11.52	7.56	8.85	7.07
[109]	10.08	12.34	12.05	13.10	18.63	21.34	15.73	13.16	14.04	18.13	13.59	12.12
[121]	13.37	10.57	6.60	7.73	7.91	11.31	14.38	14.60	12.25	12.33	11.10	11.53
[133]	10.44	9.18	11.36	17.28	16.45	15.21	12.11	12.12	14.10	14.94	21.72	16.82
[145]	12.61	13.40	12.64	12.24	9.13	12.31	19.68	19.83	15.34	15.61	14.07	13.64
[157]	16.87	11.89	12.92	19.93	23.72	23.13	13.35	11.51	18.51	17.24	11.92	12.36
[169]	13.42	11.45	11.09	14.19	14.22	12.15	10.49	11.29	11.74	9.53	7.65	7.21
[181]	7.56	8.14	11.07	16.93	11.12	8.79	16.03	18.87	17.72	14.72	14.08	14.21
[193]	9.99	6.63	10.11	12.64	15.06	14.21	14.20	16.39	16.31	16.07	17.83	20.24
[205]	14.28	17.10	17.00	18.88	17.13	23.68	24.72	19.74	24.12	33.57		